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Bihar Engineering University, Patna

B.Tech 2nd Semester Examination, 2025 (Old Course)

Code: 103201 | ***Subject: Physics (Wave & Optics and Quantum Mechanics)*** | ***Time: 03 Hours*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven question only): **[14]**

(a) Which phenomenon explains the wave nature of light?
(i) Interference
(ii) Diffraction
(iii) Polarization
(iv) All the above

(b) Which of the following is the characteristic of wave function?
(i) Continuous
(ii) Single valued
(iii) Normalizable
(iv) All of the above

(c) On reflection from denser medium what is the phase change in light wave?
(i) 180°
(ii) 90°
(iii) 135°
(iv) 0°

(d) What is metastable state?

(e) State Bragg's law of diffraction.

(f) A uniform string of length 2m and mass 200 gram is under a tension of 800N. Calculate the speed of transverse wave in the string.

(g) What is the ratio of intensities of maxima and minima in interference if the intensity ratio is 25:1?

(h) Young's double slit experiment was performed in a laboratory by taking monochromatic blue, orange and red light and fringe widths were obtained as β_B,β_O,β_R respectively. Other variables had been kept constants. Write down the fringe widths in decreasing order.

(i) The maximum velocity and maximum acceleration in a simple harmonic oscillator are numerically equal. What is the time period?

(j) State Heisenberg's uncertainty principle.

Q.2 Solve both questions : **[14]**

(a) Derive the differential equation of motion of damped harmonic oscillator. Discuss the cases of under damping, over damping and critical damping with displacement-time plots.

(b) Prove that displacement and velocity graph of SHM is elliptical.

Q.3 Solve both questions : **[14]**

(a) Discuss the construction and working of Ruby LASER. Draw its energy band diagram.

(b) What is population inversion? What is its importance in Lasing action?

Q.4 Solve both questions : **[14]**

(a) Draw a ray diagram for formation of rings in Newton's ring experiment. Derive the expressions for the diameters of nth bright and dark rings in reflected light condition.

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(b) By what factor the diameter of Newton's rings changes on using a liquid of refractive index μ in-between the glass plate and plano-convex lens.

Q.5 Solve both questions : **[14]**

(a) Explain the formation of diffraction pattern in Fraunhofer's diffraction due to a single slit. Obtain the expression for its resultant intensity. Determine the conditions for its maximum and minima.

(b) Draw the intensity distribution plot for Fraunhofer's multiple slit diffraction or a grating.

Q.6 Solve both questions : **[14]**

(a) Discuss Kronig-Penney model for the motion of an electron in a periodic potential and the existence of allowed and forbidden energy bands.

(b) Draw the total Energy (E) vs. Wave number (k) curve for an electron in 1-D lattice.

Q.7 Solve both questions : **[14]**

(a) Explain Quantum Mechanical Tunneling. Apply it to explain Alpha Decay.

(b) Find out the energy level in Sodium for which the probability of occupation at 300K is 0.50. (Given Fermi energy for Sodium is 3.13 eV. Planck's constant = 1.38×10^{-23} in S.I. Units.)

Q.8 Solve this question : **[14]**

Write down Schrödinger's wave equation for a particle inside one dimensional potential well given by:
$V(x)=0,\ 0 < x < L$
$V(x)=\infty,\ x < 0 and x > L$
Solve it to obtain energy eigen values. Show that energy eigen values are discrete.

Q.9 Write short notes on any two of the following: **[14]**

(a) Michelson's interferometer

(b) Bloch Theorem

(c) Dependence of Fermi energy on carrier concentration

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